# David Ruhé

*Curriculum Vitae*


*davidruhe.github.io*
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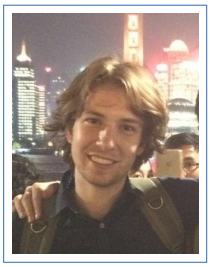
*DavidRuhe*
*25-06-1994*
*Amsterdam, The Netherlands*

## Education

2020 **PhD Candidate**, *University of Amsterdam*, Institute of Informatics, AMLab (AI4Science).

2019–2019 **MSc Thesis**, *Adversarial Attacks on Capsules*, Supervision: Maartje ter Hoeve (MSc), Prof. Zeynep Akata & Prof. Maarten de Rijke, 8.5/10.

2017–2019 **MSc Data Science**, *University of Amsterdam*, Institute of Informatics, GPA: 8.5/10 (with honors / cum laude).
Course selection: Advanced Machine Learning, Applied Machine Learning, Information Theory, Numeric Algorithms, Information Retrieval

2016 **Global Exchange Program**, Fudan University, Shanghai, China.
Courses: Elementary Chinese

2012–2016 **BSc Economics**, *University of Amsterdam*, Amsterdam School of Economics.
Course selection: Statistics 1 & 2, Econometrics, Game Theory

## Experience

2020 **PhD Candidate**, *University of Amsterdam*, Amsterdam, AMLab.
In a joint project between the machine learning lab and the astrophysics faculty I'm working on developing algorithms that can assist with detecting astronomical events in real-time. Promotors: Prof. Max Welling & Prof. Ralph Wijers. Supervision: Dr. Patrick Forré & Dr. Antonia Rowlinson

2019–2020 **Machine Learning Advisor & Researcher**, *SurfSARA*, Amsterdam.
SurfSARA is the company that runs the Dutch national supercomputer. My activities there include advising scientific projects and conducting research. I was involved in a project that seeks to generate very high-dimensional 3D voxel spaces, e.g., CT-scans.

2018–2019 **Research Intern**, *Pacmed*, Amsterdam.
At Pacmed I was part of the intensive care project studying the use of Bayesian Neural Networks and Variational Inference for better decision making at the IC unit of hospitals.

2018–2019 **Research Assistant**, *VU University*, Amsterdam.
I was accepted as a Research Assistant at the Faculty of Science at the VU univeristy Amsterdam. Together with prof. dr. Sandjai Bhulai and prof. dr. Wido van Peursen I analysed lexical variance in classical Hebrew texts using deep learning approaches.

## Programming

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|---|---|
| Intermediate | JAVASCRIPT, BASH, High Performance Computing |
| Advanced | PYTHON, implemented most common machine learning and deep learning algorithms using SKLearn, Tensorflow and PyTorch. |

## Languages

| | |
|---|---|
| Dutch | First language |
| English | Full professional fluency |
| German | Elementary proficiency. |
| Chinese | Basic words & phrases. |

# Teaching

## Courses

2021 **Machine Learning 2**, *Master AI*, University of Amsterdam.

2020 **Machine Learning 1**, *Master AI*, University of Amsterdam.

2019 **Deep Learning**, Erasmus University Rotterdam.

## Supervision

2020–2021 **Thesis Supervision - MSc**, *Yke Rusticus*, Machine Learning for Exoplanet Discovery, University of Amsterdam.

2020 **Thesis Supervision - BSc**, *Nina Verheijen*, Machine Learning for Radio Astronomy, University of Amsterdam.

# Grants

2019 **Wilmerink-Van Beeck Vollenhoven Fund**, *Student Travel Grant*.

2019 **ICML Travel Grant**.

2018 **NWO grant for access to national supercomputing facilities**, *Cartesius GPU System*.

# Awards

2019 **ICT Open**, *Meet the Demo Award*, 1st Prize.

# Publications

2021 **Self-Supervised Inference in State-Space Models** .

2021 **Detecting Dispersed Radio Transients in Real Time Using Convolutional Neural Networks** .

2019 **Generating CT-scans with 3D Generative Adversarial Networks Using a Supercomputer** , *Published at Med-NeurIPS 2019, Vancouver, BC.*.

2019 **Bayesian Modelling in Practice: Using Uncertainty to Improve Trustworthiness in Medical Applications** , *Published at ICML 2019, AISG at Long Beach CA.*.

2019 **Linguistic Variations in Classical Hebrew** .

# Conferences

2020 **Health RI 2020**, *Utrecht, Netherlands*.
Poster presentation of 3D GAN paper

2019 **NeurIPS 2019**, *Vancouver, British Columbia*, Workshop presentation of 3D GAN paper.

2019 **ICML 2019**, *Long Beach, California*, Workshop presentation of Bayesian Uncertainty paper.

2019 **ICT Open 2019**, *Hilversum, Netherlands*, Demo stand on effective traffic sign damage detection and classification.

2019 **ICT With Industry 2019**, *Lorentz Center, Leiden University*, Assisted in reating a deep-learning based pipeline that detects, and classifies traffic sign detection using 360-degree car-camera inputs, Supervisors: dr. Efstratios Gavves, dr. Massoud Mazloom.